Test Report No. 2018 – 0509

Agricultural Machinery – Hammer Mill EXCELHIGH HM-300

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Test Report Number	2018 – 0509
Date of Issuance	September 4, 2019
Valid Until	September 4, 2024
Agricultural Machinery	Hammer mill
Type	Swinging type
Brand	EXCELHIGH
Model	HM-300
Test Requested by	WINCH MECHANICAL INNOVATIONS & GENERAL CONSTRUCTION, INC. Airport Road, National Highway, Brgy. Fatima, General Santos City

SUMMARY

Construction and Performance Criteria	PAES 216:2004	AMTEC Test
Material of hammers	AISI 1080 to AISI 1085 or its ISO equivalent	Stainless steel
Fineness modulus	ND	4.01
Noise level, dB(A), maximum	96[a]	83.7[b]

[a]Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Center, Ministry of Labor, 1983.
[b]Taken at the bagger's ear level during operation with load.

OBSERVATIONS

1. Operator's manual	Not provided
2. Previous valid test reports of similar brand and model	Not provided
3. Prime mover used	1.12 kW HERCULES Electric Motor
4. Manufacturer	EXCELHIGH INDUSTRIAL TRADING AND SERVICES Brgy. Mabuhay, General Santos City
5. Test material used	Dried cassava chips
6. Loading	The machine has two input hoppers. Input hopper 1 and input hopper 2 are 1055 mm and 945 mm high from the ground respectively.
7. Cleaning of parts	Pulverizing mechanism is accessible for cleaning.
8. Adjusting and repair of pairs	Desired fineness of the product can be adjusted by changing the screen inside the milling chamber. Locally fabricated parts may be repaired.
9. Collecting of output	The milled product was collected using a basin.

OBSERVATIONS (Cont'n)

10. Transporting the machine	The machine had to be carried manually.
11. Safety	The milling shaft pulley is provided with safety cover. No safety cover was provided for the whole transmission system.
12. Vibration	Not observed
13. Others	Some of the milled product leaked through the gaps around the milling assembly housing.
14. Number of operators	Four persons: one feeding operator, two assistants in loading of input, and one bagger

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	June 8, 2018
Location of Test	Brgy. Mabuhay, General Santos City
Items Tested	Input Capacity, Output Capacity, Milling Capacity, Product Recovery, Quality of Output, Speed of Components, Power Consumption, Noise Level, and Number of Operators

METHODS OF TEST

The hammer mill was tested based on PAES 217:2004 Agricultural Machinery – Hammer Mill – Methods of Test.

DESCRIPTION OF THE MACHINE

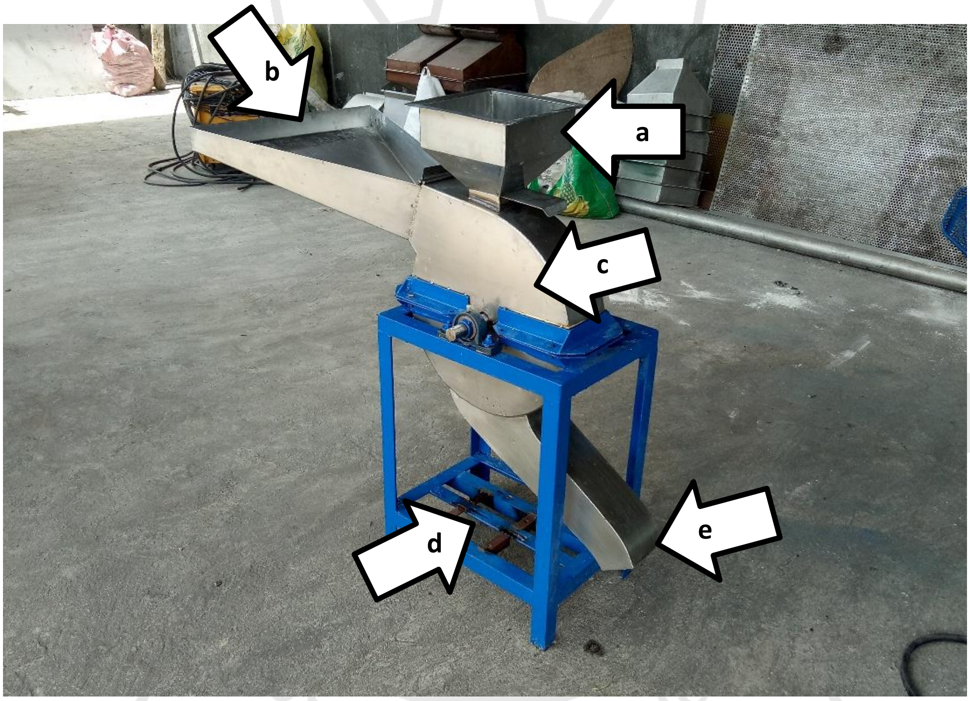

Figure 1. Parts of EXCELHIGH HM-300 Hammer mill powered by1.12 kW HERCULES Electric Motor:
(a) Input hopper 1, (b) Input hopper 2, (c) Milling chamber, (d) Prime mover base, and (e) Output chute.

SPECIFICATIONS

Item		Manufacturer's Specification[a]	AMTEC Verification[b]
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	ND	1095
1.1.2	Width	ND	510
1.1.3	Height	ND	1055
1.2	Weight without prime mover, kg	ND	NM
2	Intake hopper		
2.1	Intake hopper 1		
2.1.1	Holding capacity, kg	ND	NM
2.1.2	Height from the ground, mm	ND	1055
2.1.3	Bottom opening, L x W, mm	ND	65 x 65
2.1.4	Materials of construction	ND	Stainless steel (t = 1.4 mm)
2.1.5	Features	ND	Sliding gate valve
2.2	Intake hopper 2		
2.2.1	Holding capacity, kg	ND	NM
2.2.2	Height from the ground, mm	ND	945
2.2.3	Bottom opening, H x W, mm	ND	110 x 133
2.2.4	Materials of construction	ND	Stainless steel (t = 1.4 mm)
2.2.5	Features	ND	Stainless steel guard at inlet opening
3	Milling assembly		
3.1	Type	ND	Open
3.2	Dimension of plate, D, mm	ND	260
3.3	No. of anchor bars	ND	4
3.4	Hammer		
3.4.1	Type	ND	Swinging
3.4.2	Dimension, L x W x t, mm	ND	91 x 37 x 6
3.4.3	No. per anchor bar	ND	3
3.4.4	Means of attachment	ND	Hanging through the pins
3.4.5	Materials of construction	ND	Stainless steel

[a]Machine specifications was not given by the manufacturer.
[b]Based on on-site measurements and nameplates of machine components
Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification[a]	AMTEC Verification[b]
4	Screen		
4.1	Dimension, L x W, mm	ND	370 x 115
4.2	No. of screens	ND	1
4.3	Mesh number	ND	ND
4.4	Wire size, mm	ND	3.15
4.5	Materials of construction	ND	Stainless steel
5	Clearance between the screen and the tip of the hammer, mm	ND	4.61
6	Outlet chute		
6.1	Size of opening, H x W, mm	ND	75 x 115
6.2	Height from the ground, mm	ND	210
6.3	Materials of construction	ND	Stainless steel (t = 1.4 mm)
7	Aspirator	NA	NA
8	Primemover	ND	Electric motor
8.1	Brand	ND	HERCULES
8.2	Type	ND	Induction
8.3	Make or manufacturer	ND	HERCULES Electric Co. Ltd. Taiwan
8.4	Serial number	ND	ND
8.5	Rated power, kW	ND	1.12
8.6	Rated speed, rpm	ND	ND
8.7	Phase	ND	Single
8.8	Voltage, V	ND	220
8.9	Rated maximum current, A	ND	ND
8.10	Frequency, Hz	ND	60
9	Safety devices	ND	, No belt cover, cover for milling shaft pulley only
10	Special features	ND	None

[a]Machine specifications was not given by the manufacturer.
[b]Based on on-site measurements and nameplates of machine components
Note: ND – No Data, NM – Not Measured, NA – Not Applicable

RESULTS

Item		Data	
1	Conditions of Test Sample		
1.1	Crop	Dried cassava chips	
1.2	Variety	ND	
1.3	Source	Brgy. Mabuhay, General Santos City	
1.4	Moisture content, $\%_{wb}$	12.8	
2	Weight of input, kg	31.8	
3	Input time, h	0.183	
4	Weight of output, kg	30.0	
5	Output time, h	0.216	
6	Operating time, h	0.217	
7	Input capacity, kg/h	173.4	
8	Output capacity, kg/h	138.7	
9	Milling capacity, kg/h	138.1	
10	Product recovery, %	94.3	
11	Speed of components, rpm	Without load	With load
11.1	Electric motor	1785	1780
11.2	Pulverizer cylinder shaft	1876	1778
12	Noise level, dB(A)		
12.1	Feeding operator	72.9	82.6
12.2	Bagger	73.1	83.7
13	Power consumption		
13.1	Voltage, V	227.9	226.0
13.2	Current, A	3.88	5.54
13.3	Power, kW	0.89	1.22
14	Number of operators	Four persons: one feeding operator, two assistants in loading of input and one bagger	

Note: ND – No Data

RESULTS (Cont'n)

Item		Data
15	**Laboratory analysis**	
15.1	US standard sieve number	Weight fraction retained, %
15.1.1	#12 (1.680 mm)	11.04
15.1.2	#16 (1.191 mm)	36.84
15.1.3	#20 (0.841 mm)	19.93
15.1.4	#30 (0.594 mm)	15.37
15.1.5	#40 (0.420 mm)	10.14
15.1.6	#50 (0.297 mm)	4.85
15.1.7	Pan	1.84
15.2	Fineness Modulus	4.01
15.3	Average particle size diameter, mm	0.989

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SEP 04 2019
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